

Power via Barrel Jack/USB

The Arduino needs a source to power it up. There are two ways to power it: you can do it either via the USB or the Barrel Jack. You can connect the USB to a laptop/computer to power up the board. There is a sort of regulator present which prevents voltage from overflowing via the USB. The power is generally supplied at 100mA @ 5V via the USB. You can also choose to furnish the barrel jack a voltage of 7V-12V. Another regulator is present to maintain the power supply and to prevent overflow.



<https://www.amazon.com>

Reset Button

The Arduino needs a Reset button to restart any code that has been running on it. Clicking on the Reset button will connect the reset pin to the ground and restart the code for you to begin fresh. It doesn't help in overcoming the error in your code but allows you to start again. Whenever you trigger the reset button, the microcontroller will begin from 0.

LED Indicator

Also present is a tiny LED indicator which indicates if the device is ON or OFF. If there is any fault with the circuit, the LED is one of the prime indicators of the same.

Voltage Regulator

The voltage regulator does what its name suggests, maintains and manages how much power reaches the Arduino circuit. Like any other voltage regulating components, it protects the devices from the potential shock of voltage outburst. It is the stabilizing agent that supplies the required 5V heat to the board, irrespective of how much heat it gets from the other end.

RX TX LEDs

RX defines 'receive' whereas TX represents 'transmit.' These are the pins responsible for serial communication. The respective LED lights up when-